

Task 1 - Product Management REST API

Objective:

Develop a REST API using Spring Boot for Product Management with CRUD operations.

Technologies Used:

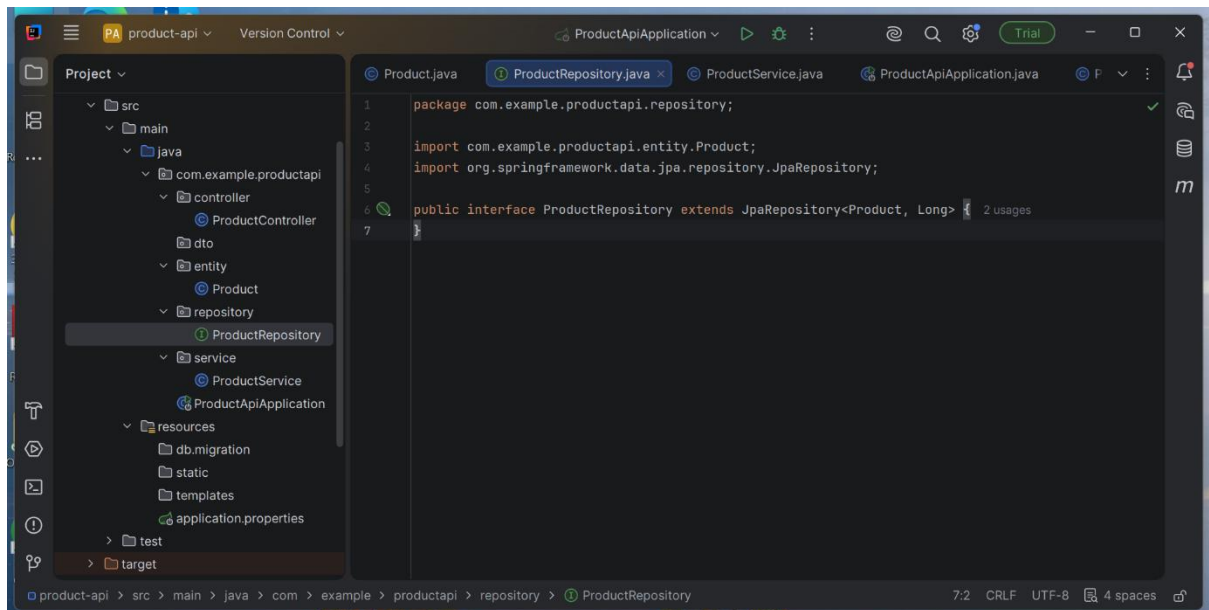
- > Java 17
- > Spring Boot
- > Spring Data JPA
- > MySQL
- > Maven

Features Implemented:

- > Create Product
- > Get All Products
- > Get Product By ID
- > Update Product
- > Delete Product

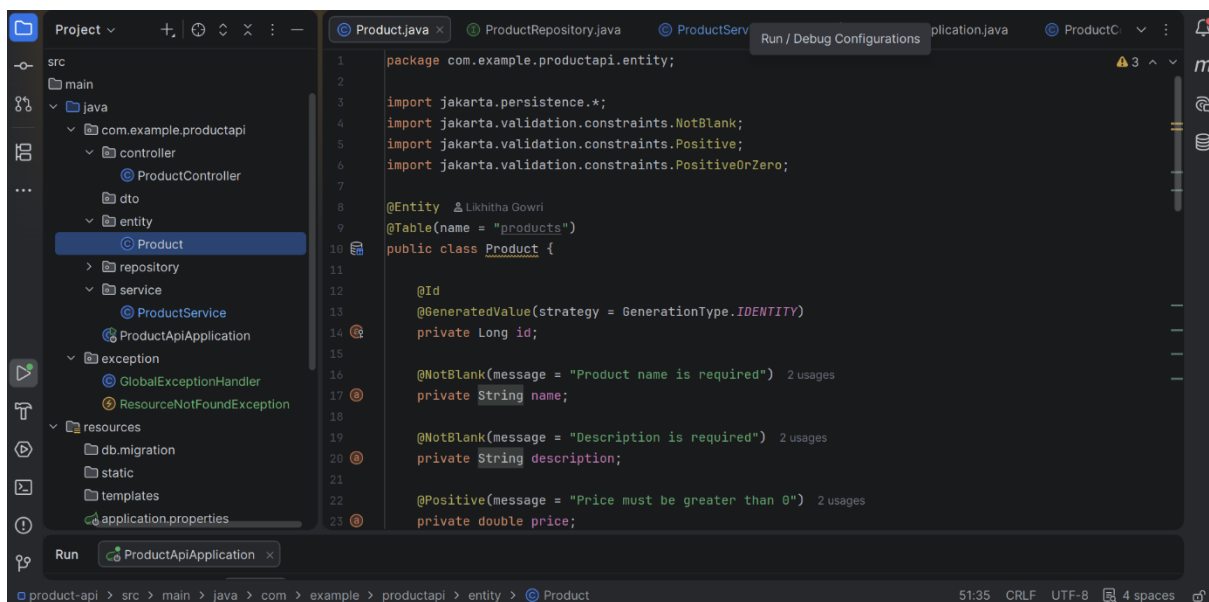
Project Structure

This project follows a layered architecture using Controller, Service, Repository, and Entity classes



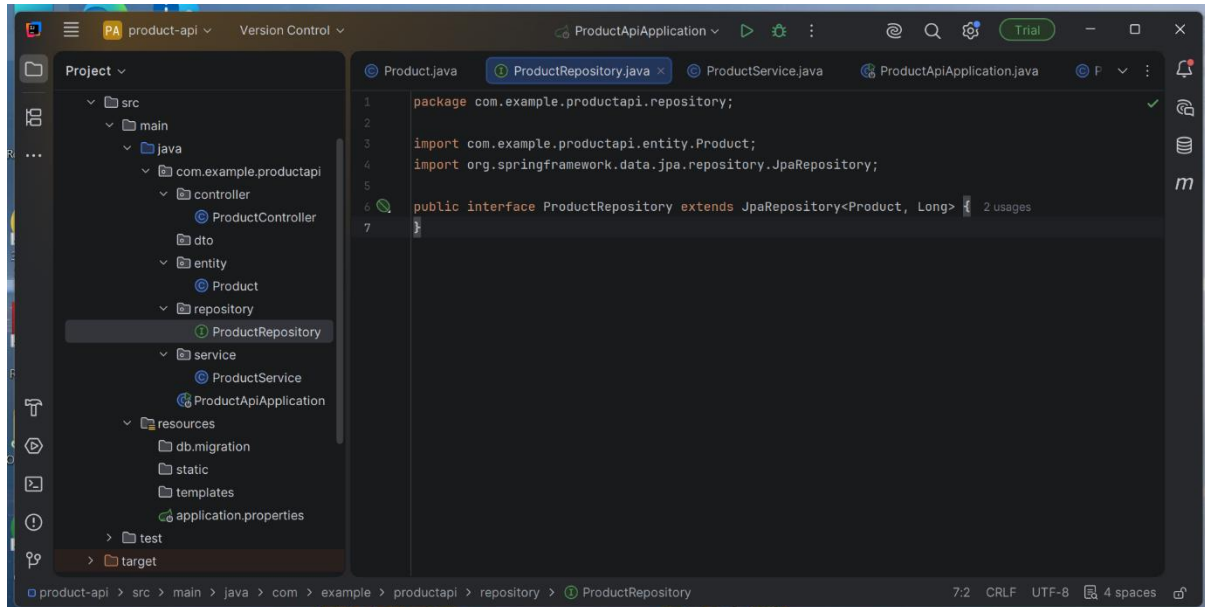
Product Entity

The Product entity represents product information such as name, description, price, and quantity



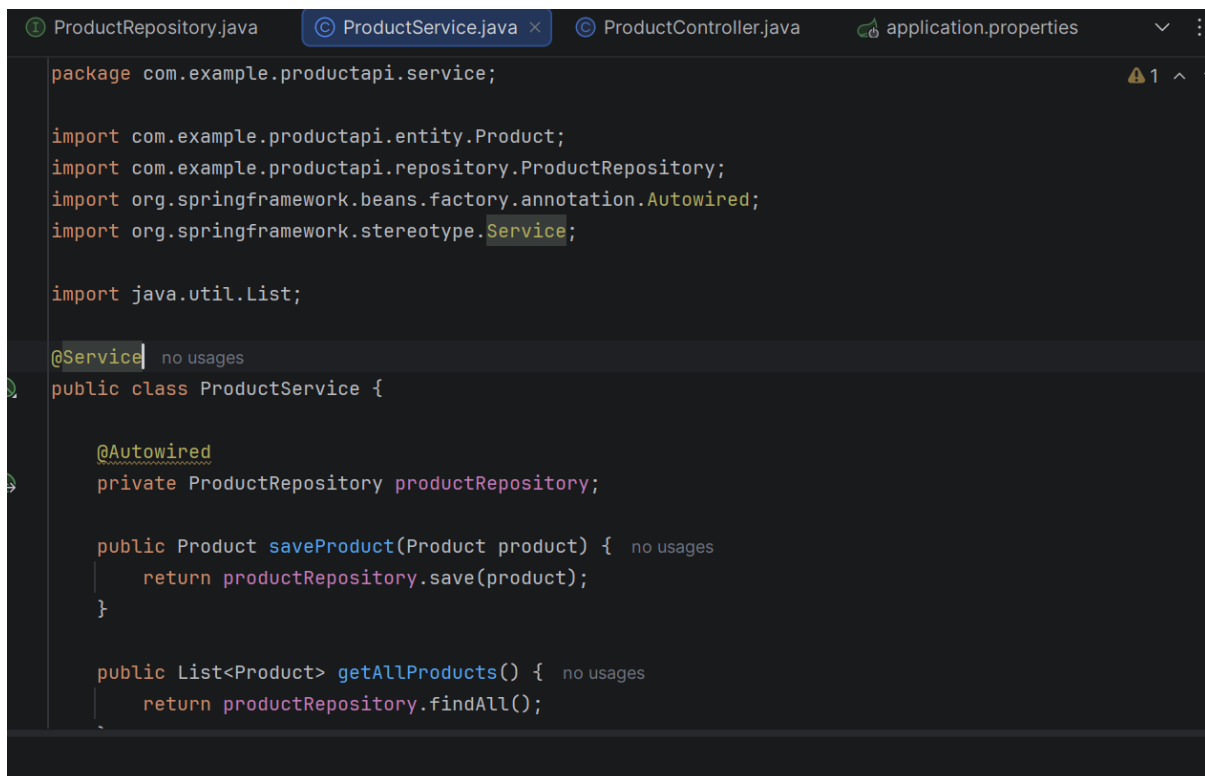
Repository Layer

The repository layer uses Spring Data JPA to perform database operations.



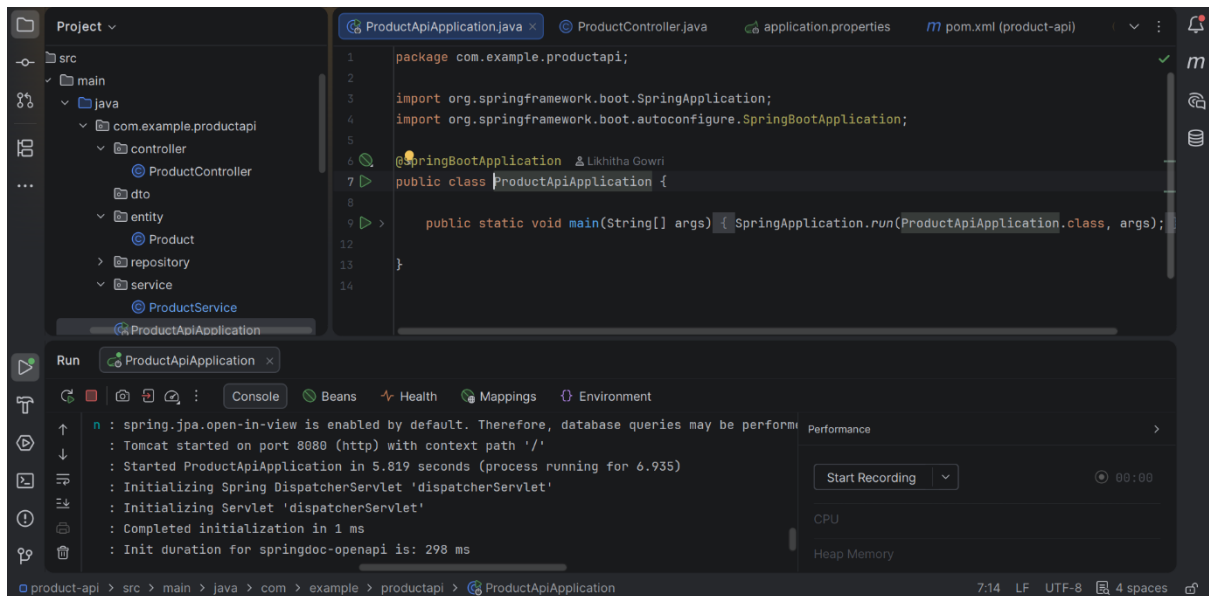
Service Layer

The service layer contains business logic for CRUD operations.



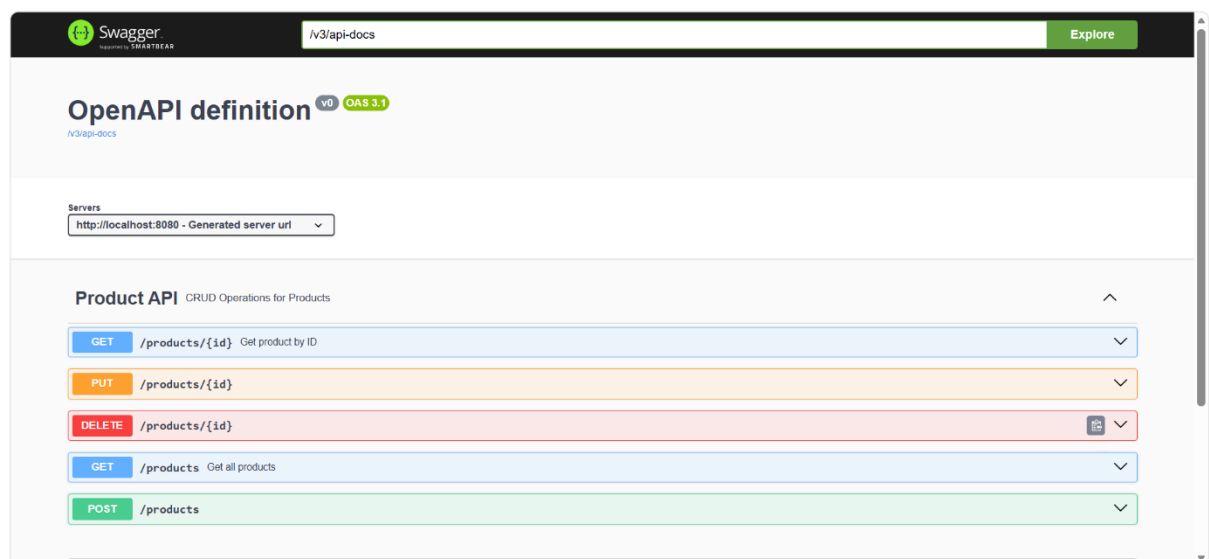
Application Execution

The Spring Boot application was executed successfully.



Swagger API Documentation

Swagger UI was integrated to test and document REST APIs.



Get All Products API

Successfully retrieved product data from the database.

GET /products Get all products

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8080/products' \
  -H 'accept: */*'
```

Request URL

http://localhost:8080/products

Server response

Code

Details

200

Response body

```
{
  "description": "HP Victus",
  "id": 4,
  "name": "Gaming Laptop",
  "price": 65000,
  "quantity": 5
}
```

Download

Response headers

GET /products/{id} Get product by ID

Parameters

Name	Description
id * required	
integer(\$int64)	4
(path)	

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8080/products/4' \
  -H 'accept: */*'
```

Request URL

http://localhost:8080/products/4

Server response

Code

Details

200

Response body

```
{
  "description": "HP Victus",
  "id": 4,
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GitHub Repository

Project Source Code:

<https://github.com/likhithagowri909/product-api>

Result

Successfully developed and tested a Product Management REST API using Spring Boot, Spring Data JPA, and MySQL. Implemented complete CRUD operations and integrated Swagger UI for API documentation and testing.